

E-COMMERCE CONVERSION DYNAMICS: CUSTOMER BEHAVIOR & PURCHASE INTENT ANALYSIS

Customer Behavior, Conversion
Dynamics & Purchase Intent



TOOLS USED

Python, Pandas, Numpy, Matplotlib, Plotly &
ML libraries

PRESENTED BY

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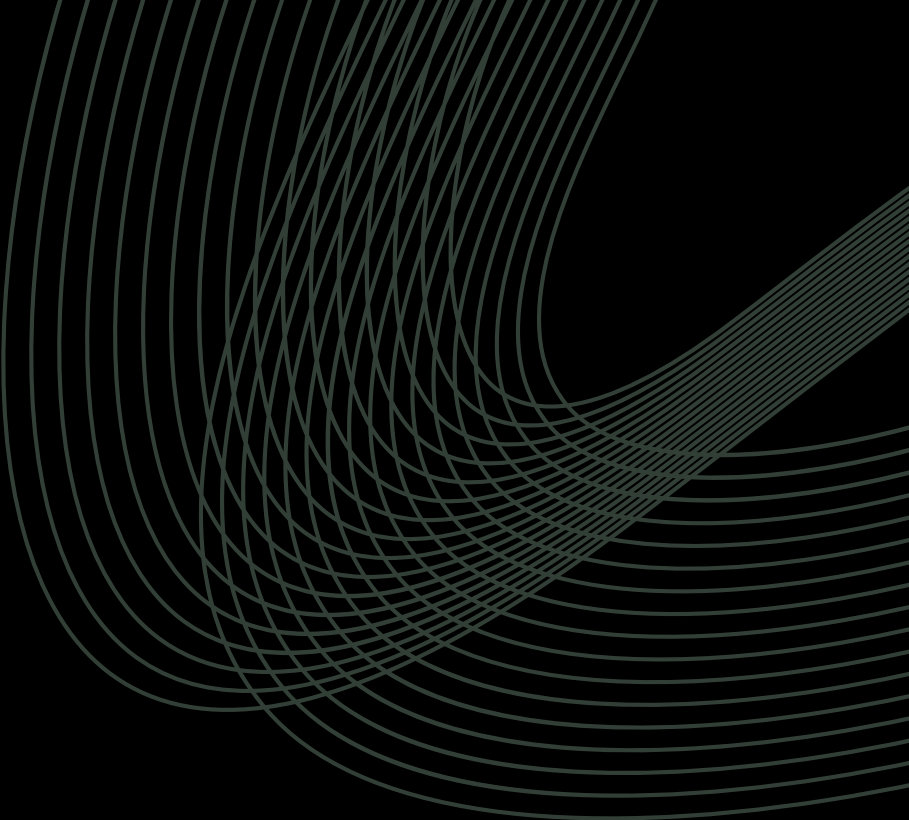
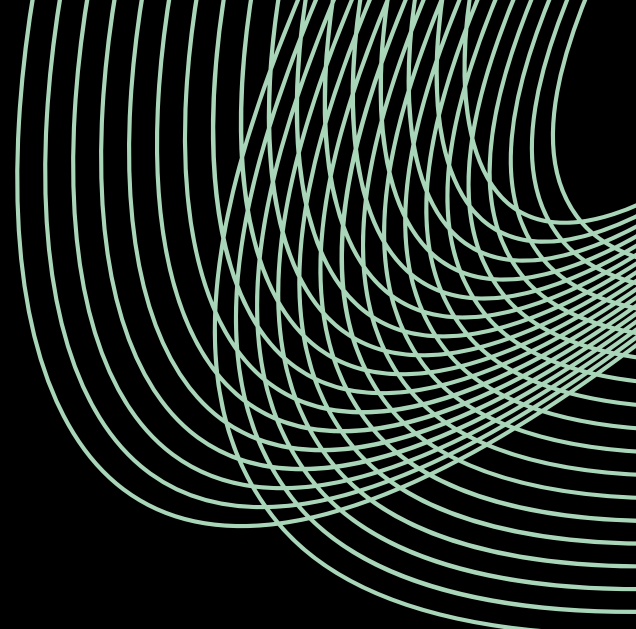


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Overview & Business Objective



1.1 PROJECT OVERVIEW

E-commerce platforms operate in a dynamic environment where purchase decisions are influenced by product engagement, traffic sources, visitor type & session behavior.

Understanding these behavioral patterns helps businesses identify sessions associated with stronger purchase conversion and better understand online customer activity.

The E-Commerce Conversion Dynamics project uses session-level online shopping data and machine learning to analyze purchase behavior and predict whether a session results in a completed purchase.

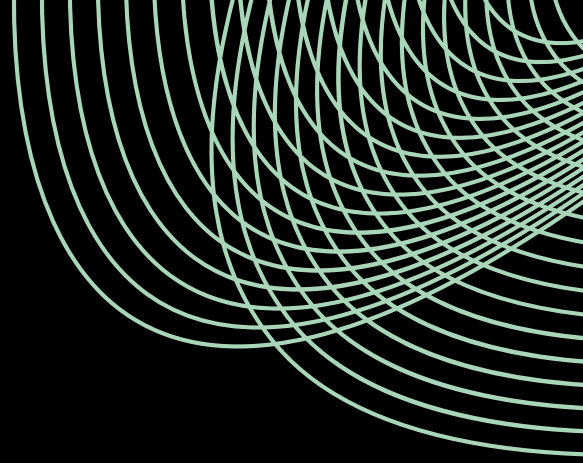
The project combines:

- Data understanding and cleaning
- Exploratory Data Analysis
- Behavioral feature engineering
- Class imbalance handling
- Machine learning classification
- Model evaluation and comparison
- Prediction-error analysis
- Model explainability
- Business-oriented insights

The objective is not only to predict purchase conversion accurately, but also to understand the behavioral signals behind the model's predictions and how these insights can support better e-commerce decision-making.



1.2 PROBLEM STATEMENT



Purchase conversion is influenced by multiple interacting aspects of an online shopping session, making it difficult to identify potential purchases using individual behavioral indicators alone.

The key problem addressed in this project is:

How can session-level e-commerce data be used to build a machine learning model that accurately identifies purchase sessions while also understanding the key behavioral factors associated with conversion?

The project therefore treats **Revenue** as the target variable and uses browsing behavior, product engagement, page value, traffic sources, visitor characteristics, and technical session information as predictive features.



1.3 PROJECT OBJECTIVES

The major objectives of the project are:

- Understanding the structure, quality, and characteristics of the online shopping dataset.
- Analyzing purchase conversion patterns across months, visitor types, traffic sources, and session behavior.
- Identifying relationships between browsing activity, product engagement, and purchase outcomes.
- Creating meaningful behavioral features and preparing the dataset for machine learning.
- Addressing class imbalance and developing multiple classification models for purchase prediction.
- Tuning and comparing classification models using appropriate performance metrics.
- Evaluating classification thresholds and investigating false-positive and false-negative predictions.
- Explaining model predictions using feature importance and permutation importance.
- Translating analytical and modeling results into practical e-commerce business insights.



1.4 PROJECT SCOPE

The project follows a structured workflow covering data analysis, behavioral feature engineering, machine learning, model evaluation, and explainability to understand purchase patterns, identify key conversion signals, evaluate predictive performance, and derive actionable e-commerce insights.

01

Data Analysis

- Dataset inspection
- Data quality assessment
- Purchase conversion patterns
- Behavioral relationship analysis

02

Conversion Modeling

- Feature preparation
- Behavioral feature engineering
- Classification modeling
- Model tuning and comparison

03

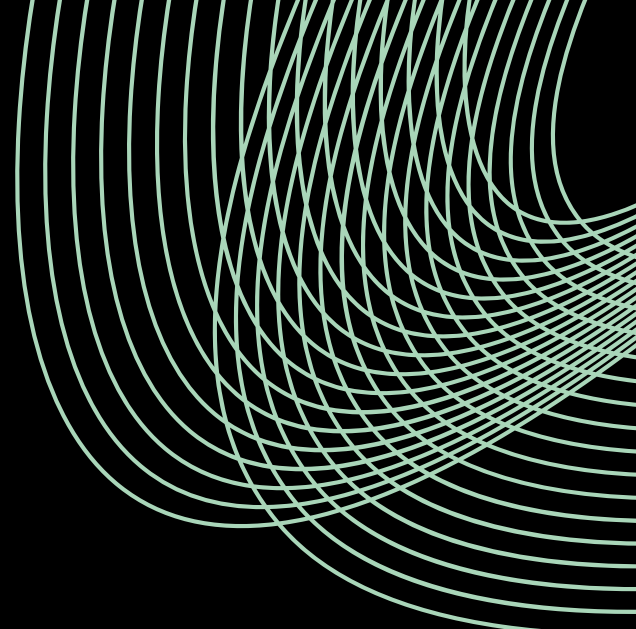
Explainability

- Feature importance analysis
- Original-feature aggregation
- Permutation importance analysis

04

Business Interpretation

- Purchase conversion observations
- Customer behavior insights
- Conversion signal interpretation
- Recommendations for e-commerce optimization



Data Understanding & Exploratory Data Analysis



2.1 DATA OVERVIEW

The dataset provides a comprehensive view of e-commerce shopping sessions, covering browsing activity, page engagement, visitor characteristics, technical attributes, traffic sources, special-day proximity, and purchase outcomes. These variables provide the foundation for analyzing customer behavior, identifying factors associated with purchase conversion & developing predictive models for e-commerce purchase intention.

Data Source: UCI

Dataset Size

12,330

Records

18

Original Features

Purchase Diversity

9

Regions

3

Visitor Type

20

Traffic Type

